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| **Fiscal Year:** | FY 2015 | **Task Last Updated:** | FY 01/28/2016 |
| **PI Name:** | Zhang, Ye M.D., Ph.D. | | |
| **Project Title:** | Interphase Chromosome Conformation and Chromatin-Chromatin Interactions in Epithelial Cells Cultured Under Different Gravity Conditions | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | SPACE BIOLOGY | | |
| **Program/Discipline--Element/Subdiscipline:** | SPACE BIOLOGY--Cellular and molecular biology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | (1) Cell Culture | | |
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| **Organization Name:** | NASA Kennedy Space Center | | |
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| **City:** | Kennedy Space Center | **State:** | FL |
| **Zip Code:** | 32955 | **Congressional District:** | 8 |
| **Comments:** | | | |
| **Project Type:** | Ground,New Investigation | **Solicitation / Funding Source:** | 2012 Space Biology NNH12ZTT001N |
| **Start Date:** | 10/01/2013 | **End Date:** | 03/31/2015 |
| **No. of Post Docs:** | | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA ARC |
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| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: Extended to 3/31/2015, per A. Chu/ARC (Ed., 8/6/14) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Wu, Honglu ( NASA Johnson Space Center )<br>Hada, Megumi ( NASA-JSC/Wyle laboratories ) | | |
| **Grant/Contract No.:** | Internal Project | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | Microgravity has been shown to alter global gene expression patterns and protein levels in cultured cells or animals. It has been suggested that packaging of the chromatin fibre in the interphase nucleus is closely related to genome function, and the changes in transcriptional activity are tightly correlated with changes in chromatin folding. Several principles of chromatin folding have been widely accepted and have revealed that an extensive degree of folding of human 46 chromosomes. In an interphase nucleus of about 5-20 μm in diameter, a total length of 5 cm of nucleosomal filaments is packaged as beads being aligned on a string. Furthermore, data obtained from various molecular technological approaches by various research groups, including our own team, have provided more and more evidences that local chromatin folding is related to the transcriptional activity and gene density. However, there is no data available currently on chromosomal conformation changes in micro-G, which may extensively influence cellular function, differentiation and development, due to the subsequent regulation of gene expressions caused by altered chromatin folding. This study explores the changes of chromatin conformation and chromatin-chromatin interactions in the simulated microgravity and super-G environments, and their correlation to the changes in the expression of genes located at different regions of the chromosome. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | It was the first time that mBAND painting technique was applied to determine the three-dimensional structure of chromatin folding using confocal imaging acquisition. The findings on the non-random chromatin folding in epithelial cells indicate that the changes of chromosome conformation may alter the break-point distribution in radiation-induced intrachromosomal exchanges, which may subsequently alter the long-term outcome produced by radiation exposure. |
| **Task Progress:** | FINAL REPORT PROGRESS REPORTED JANUARY 2016:<br>In this project, we concluded that on a multi-megabase pair scale of DNA, the arrangement of chromatin in M10 epithelial cells was found to be non-random. In human epithelial cells, both telomere regions tended to be located towards the exterior of the chromosome domain, whereas the remaining p-arm chromatin region was located towards the interior. In contrast, most of the q-arm of the chromatin was found in the periphery of the domain. Chromosome conformation in a significant number of cells was found to be altered under a simulated microgravity condition. The alterations included rounder shape of the nucleus, poor correlations of two homologous chromosomes and their regions, and changes in the relative location of region 2 that contains the 3p21 hotspot in a significant number of chromosome 3s.<br><br>ANNUAL PROGRESS REPORTED AUGUST 2014: We revisited the data obtained from a preliminary study on the effects of proximity within interphase chromosome on the breakpoint distribution in radiation-induced intra-chromosomal exchanges and completed data analysis. The data indicates that the telomere regions as well as the q-arm of the chromosome appeared to be located towards the exterior of the chromosome domain. Most of the regions in the p-arm of the chromosome as well as the centromere region, however, were closer to the center of the chromosome domain. In addition, the interior of the chromosome domain was preferentially occupied by the p-arm of the chromatin, which is consistent with our previous finding of intrachromosome exchanges involving breaks on the p-arm and in the centromere region of chromosome3. This work was recently published in a peer-reviewed journal.<br><br>To simulate microgravity on the ground, we evaluated the Random Positioning Machine (RPM) and the available HARV bioreactors and decided to acquire a new RPM to model different gravity conditions. RPM allows simulation of microgravity in adherent cells cultured cells in 2D. Another important reason is that it is possible to use RPM to simulate various gravity conditions. The process to acquire RPM, including the market survey and procurement from a European company, took several months. The system was successfully tested and is being used for various projects. In this study, the epithelial cells were growing from 20% confluence for five days to reach confluence under both RPM and static culture conditions. No significant changes of morphology, growth rate, and cell death were found.<br><br>To determine whether the pattern of interphase chromatin folding is cell type specific, we analyzed the 3D structure of interphase chromosome 3 in G0 lymphocytes. The preliminary result suggested that the folding of interphase chromosome 3 may be cell type specific. In comparison to the chromosome in static epithelial cells, the interphase structure of the chromosome in epithelial cells cultured in the simulated microgravity condition for 5 days appeared to be different. Total RNAs have been isolated for global gene array analysis. The analyses are time-consuming and in progress.<br><br>Future work for the final year of the project: • Complete the gene array analysis and determine the correlation of gene expression changes and the changes of interphase chromatin folding in chromosome 3; • Complete the chromosome aberration analysis to determine the relationship between the distribution of intrachromosomal exchanges and the changes of interphase chromatin folding in chromosome 3 under simulated microgravity environment; • Prepare a full proposal to explore the changes of chromatin folding in cells during actual spaceflight, which may further explore other chromosomes and/or cell types. |
| **Bibliography Type:** | Description: (Last Updated: 01/08/2024) |
| **Abstracts for Journals and Proceedings** | Zhang Y, Hada M, Wu H. "Interphase Chromosome Conformation and Chromatin-chromatin Interactions in Human Epithelial Cells Cultured under Different Gravity Conditions." Oral presentation in 3rd Annual International Space Station (ISS) Research and Development Conference, Chicago, Illinois, June 17-19, 2014.<br>3rd Annual International Space Station (ISS) Research and Development Conference, Chicago, Illinois, June 17-19, 2014. , Jun-2014 |
| **Abstracts for Journals and Proceedings** | Zhang Y, Hada M, Wu H. "Interphase Chromosome Conformation and Chromatin-chromatin Interactions in Human Epithelial Cells Cultured under Different Gravity Conditions." Oral presentation in the 30th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), Pasadena, CA, October 22-26, 2014.<br>30th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), Pasadena, CA, October 22-26, 2014. , Oct-2014 |

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| **Abstracts for Journals and Proceedings** | Zhang Y, Uhlemeyer J, Hada M, Asaithamby A, Chen DJ, Wu H. "Proximity within interphase chromosome contributes to the breakpoint distribution in radiation-induced intrachromosomal exchanges." 15th International Congress of Radiation Research, Kyoto, Japan, May 25-29, 2015.<br>15th International Congress of Radiation Research, Kyoto, Japan, May 25-29, 2015. , May-2015 |
| **Abstracts for Journals and Proceedings** | Zhang Y, Wong M, Hada M, Feiveson AH, Wu H. "Interphase Chromosome Conformation in Mammalian Cells Cultured under Simulated Microgravity Conditions." 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015.<br>31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. , Nov-2015 |
| **Articles in Peer-reviewed Journals** | Zhang Y, Moreno-Villanueva M, Krieger S, Ramesh GT, Neelam S, Wu H. "Transcriptomics, NF-kappaB pathway, and their potential spaceflight-related health consequences." Int J Mol Sci. 2017 May 31;18(6):1166. Review. https://doi.org/10.3390/ijms18061166 ; PubMed PMID: 28561779 ; PubMed Central PMCID: PMC5485990 , May-2017 |
| **Articles in Peer-reviewed Journals** | Zhang Y, Story M, Yeshitla S, Wang X, Scully RR, Theriot C, Wu H, Ryder VE, Lam CW. "Persistent changes in expression of genes involved in inflammation and fibrosis in the lungs of rats exposed to airborne lunar dust." Inhal Toxicol. 2023 Mar 26;1-18. https://doi.org/10.1080/08958378.2023.2172485 ; PMID: 36966416 , Mar-2023 |
| **Articles in Peer-reviewed Journals** | Zhang Y, Uhlemeyer J, Hada M, Asaithamby A, Chen DJ, Wu H. "Proximity within interphase chromosome contributes to the breakpoint distribution in radiation-induced intrachromosomal exchanges." Life Sciences in Space Research. 2014 Jul;2:23-8. http://dx.doi.org/10.1016/j.lssr.2014.05.004 , Jul-2014 |
| **Awards** | Zhang Y, Uhlemeyer J, Hada M, Asaithamby A, Chen DJ, Wu H. "Excellent Poster Award for 'Proximity within interphase chromosome contributes to the breakpoint distribution in radiation-induced intrachromosomal exchanges.' 15th International Congress of Radiation Research, Kyoto, Japan, May 25-29, 2015." May-2015 |